

Samarth Vijay Bhamare

Agentic AI Engineer · Software Engineer (Backend) · LLM Systems

bhamaresamarth@gmail.com | +91 9075002770 | Pune, India | linkedin.com/in/samarth-bhamare | github.com/Samarth0211

PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience designing and delivering production-grade **Agentic AI** systems, **scalable backend APIs**, and **distributed data pipelines**. Sole architect of MedXprts — a **multi-agent LLM** platform that cut healthcare report turnaround from 3–4 days to under 1 hour, automating 400+ monthly workflows for a US-based client. Expertise spans the full stack: **tool-calling**, **RAG**, **LLM evaluation**, **RESTful API design**, and **serverless architecture on AWS**. M.Sc. Data Science, Deakin University, Australia.

WORK EXPERIENCE

DataKernels Analytics & Consulting LLP

Apr 2024 – Present

Software Engineer

► MedXprts — Agentic AI & Document Intelligence Platform *LangChain · FastAPI · AWS · Snowflake*

- **96% faster delivery (3–4 days → <1 hr)**: Sole architect of end-to-end **Agentic AI** platform that ingests 1,500+ page patient document packages and autonomously generates **10-section Life Care Plan (LCP) reports** — covering medical history, future care needs, and cost projections — for a US medicolegal client at 400+ workflows/month.
- **95% accuracy vs. manual baseline**: Engineered **LangChain** multi-agent pipeline — specialist agents extract diagnoses, treatment history, and cost data from OCR'd records via **RAG** and **tool-calling**; **reflection loops** self-validate each LCP section against CRM ground-truth before persisting to **Snowflake** and MongoDB.
- **153-point LCP quality framework**: Built per-section **hallucination-detection** and **production quality monitoring** system across all 10 LCP sections using **BLEURT**, **BERTScore**, **ROUGE-L**, and **semantic similarity** — severity-tiered alerts flag critical medical inaccuracies before report delivery.
- **85% latency reduction (90–120s → 12–20s)**: Redesigned **FastAPI** backend to generate all 10 LCP sections in **parallel** — each section routed to a dedicated **LLM** agent concurrently — resolving **concurrency** bottlenecks that previously forced sequential section processing.

► FeeViewer — System Reverse Engineering & Backend Automation *Python · AWS · CLI*

- **10x speedup (5–7s → 0.5s)**: Reverse-engineered a GUI-only third-party tool into a **programmatic CLI service** on AWS — resolved **COM threading** violations, exposed a **REST-compatible interface**, and enabled full **automated pipeline integration**.

► Data Infrastructure & Workflow Orchestration *AWS · Redshift · n8n · S3*

- **Scaled to 500K+ rows**: Designed **distributed data migration** from Airtable to **AWS Redshift Serverless**; built **serverless ETL pipelines** using Lambda and S3 with fault-tolerant retry logic.
- **400+ automated workflows**: Architected **event-driven orchestration** using **n8n** managing **microservice** dependencies across AWS, external **REST APIs**, and internal data stores.
- **Led 4 engineers** post-launch on front-end and case-tracking delivery — independent modules shipped within 6 weeks.

Business Analyst

► Business Intelligence Platform *Power BI · Snowflake · AWS · SQL*

- **Full BI stack, zero to production**: Designed end-to-end **data pipeline (AWS S3 → Snowflake → Zoho CRM)** and built **Power BI (DAX)** dashboards tracking physician volumes, referral performance, and case outcomes for a US client.
- **SQL-driven KPI reporting**: Authored complex **Snowflake SQL** queries consolidating 3 data sources into a single **optimised reporting pipeline**, eliminating manual reporting.

PROJECTS

CloudPilot · *Multi-Cloud Cost Intelligence SaaS* github.com/Samarth0211/CloudPilot

- **AI agent** aggregates AWS/GCP/Azure spend; delivers context-aware **cost optimisation** and infra troubleshooting grounded in the user's live environment via read-only **API access**. Stack: **Python, FastAPI, LangChain, Snowflake, Vertex AI**.

AI Madesy · *AI Automation YouTube Channel* | myrashifal.in · *Freemium Web Application*

- **3,000+ subscribers** on **LLM workflows** and AI tooling. Independently built and operate myrashifal.in — live freemium web app demonstrating full-stack product ownership.

EDUCATION

Master of Data Science · GPA 3.58 / 5.0

Jul 2023 – Oct 2025

Deakin University · Melbourne, Australia

- **Capstone — NLP Recommendation Engine:** Built **transformer-based + collaborative-filtering** model using **NLP** on watch history and YouTube comment sentiment; achieved 87% precision@10 on held-out test set.

B.E. Computer Engineering · CGPA 8.75 / 10.0

Mar 2021 – May 2023

Savitribai Phule Pune University · India

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, Agentic Workflows, Tool-Calling, RAG, LLM Orchestration, Prompt Engineering, Reflection Loops, Hallucination Detection

LLM Evaluation: BLEURT, BERTScore, ROUGE-L, Semantic Similarity, Production Quality Monitoring

Backend & System Design: Python, FastAPI, REST API Design, Microservices, Distributed Systems, Concurrency, Serverless Architecture, Flask

Cloud & Infrastructure: AWS (Lambda, S3, EC2, Redshift Serverless), GCP, Vertex AI, Docker, ETL Pipelines

Data & Analytics: Snowflake, MongoDB, PostgreSQL, SQL, Power BI (DAX), n8n, Workflow Orchestration

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300) — Microsoft
- Process Data from Dirty to Clean — Google
- National Level Hackathon Winner — ML model optimisation & fine-tuning — Great Learning · Aug 2023